Name:

Roll No:

RCSS
RAJAGIRI COLLEGE OF SOCIAL SCIENCES

Rajagiri College of Social Sciences (Autonomous)
Continuous Assessment Examination - II
October 2025
I MCA

Code: MCA104 **Total Time : 60 minutes**

Sub: Computer Networks and Internet Architecture **Total Mark : 20**

Level	Bloom's Taxonomy Levels of Learning
L1	Remembering
L2	Understanding
L3	Applying
L4	Analyzing
L5	Evaluating
L6	Creating

SECTION A
Each question has 10 marks

Sl.no	Question	CO Mapped	Bloom's Taxonomy level
1A	A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x3+1. a. What is the actual bit string transmitted? Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?	CO2	L3
	OR		
1B	A LAN has 5 computers (A, B, C, D, E) connected using CSMA/CD. The time it takes a signal to travel between any two computers is 25 μs. Each frame takes 100 μs to send. Scenario: Computer A starts sending a frame at the same time as computer C. a) What happens when A and C send at the same time? b) How do the computers know that a collision has happened? c) What do the computers do after detecting the collision?	CO3	L3

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SECTION B

Each question has 10 marks

2A	A company is setting up a small network to support its employees' web browsing, email, and file transfers. The network has the following requirements: • Employees need to browse websites on the internet. - HTTP • Employees need to send and receive emails. SMTP • Employees need to upload and download large project files from a central server. FTP a) Identify which protocol should be used for each requirement. b) For the file transfer requirement, explain how FTP modes (active and passive) affect connectivity if employees are behind a firewall. c) If an employee tries to send an email but it doesn't reach the recipient, explain how you would troubleshoot using SMTP concepts. d) Describe how HTTP handles multiple simultaneous requests when employees browse different websites at the same time.	**CO4**	**L3**
	OR		
2B	Discuss the transport layer protocols 1. Go Back N 2. Selective Repeat 3. Piggybacking	**CO4**	**L2**

Name: Drija
Reg. No.:

Rajagiri College of Social Sciences (Autonomous)

First Semester MCA Degree Examination

October, 2025

(Regular - 2025 admission)

Code: 6037 **Max. Marks: 75**

Sub: (MCA104) Computer Networks and Internet Architecture **Duration: 3 Hrs.**

SECTION A

*Answer **ALL** questions.*

Each question carries 15 marks

[5 X 15 = 75]

1 (a) i) a) Define multiplexing.

b) Explain FDM and TDM with neat diagrams. (9 Marks)

ii) Define data communication and list its fundamental components. (3 Marks)

iii) Describe the concept of framing in data link control. (3 Marks)

[OR]

(b) i) a) Explain the working principle of the Go-Back-N ARQ protocol.

b) Compare it with the Selective Repeat ARQ protocol in terms of efficiency. (9 Marks)

ii) Elaborate the role of flow control in the data link layer. (3 Marks)

iii) How does Time Division Multiplexing (TDM) differ from Frequency Division Multiplexing (FDM)? (3 Marks)

2 (a) i) a) Explain how ICMP works in network troubleshooting.

b) Illustrate with examples of echo request and destination unreachable messages. (9 Marks)

ii) What are the primary responsibilities of the network layer? (3 Marks)

iii) Explain how IPv6 simplifies packet processing compared to IPv4. (3 Marks)

[OR]

(b) i) a) Explain the functions of the network layer in detail.

b) Describe the services it provides to the transport layer. (9 Marks)

ii) Analyze the need for routing tables and how they are updated in dynamic routing. (3 Marks)

iii) What is meant by subnetting? (3 Marks)

2

P.T.O

3 (a) i) a) Discuss the major flow control mechanisms in TCP.

b) Explain how acknowledgments and timers are used to maintain flow control.

(9 Marks)

ii) What is the difference between connection-oriented and connectionless services? (3 Marks)

iii) What does QoS stand for? Mention any two of its key parameters. (3 Marks)

[OR]

(b) i) a) Compare the use cases of TCP and UDP in real-time applications.

b) Evaluate why some applications prefer UDP despite its unreliability.

(9 Marks)

ii) Describe how congestion occurs in a network. (3 Marks)

iii) Discuss how UDP achieves low latency communication despite its unreliability. (3 Marks)

4 (a) i) a) Define cryptography and list its main goals.

b) Differentiate between plaintext, ciphertext, and key.

(9 Marks)

ii) What is the main function of the Application Layer in the OSI model? (3 Marks)

iii) Describe the main objective of SSL/TLS protocols. (3 Marks)

[OR]

(b) i) a) Discuss the vulnerabilities associated with HTTP and FTP.

b) Propose secure alternatives and explain how it mitigates risks.

(9 Marks)

ii) Explain how firewalls filter packets based on security rules. (3 marks)

iii) Explain how DNS resolves a domain name to an IP address step by step.

(3 Marks)

5 (a) i) a) Evaluate the role of cloud computing in modern Internet service delivery.

b) Discuss its impact on scalability and cost efficiency for enterprises.

(9 Marks)

ii) What is the Internet, and how is it different from an intranet? (3 Marks)

iii) Differentiate between Tier-1, Tier-2, and Tier-3 ISPs. (3 Marks)

[OR]

(b) i) a) Define an ISP and explain its role in providing Internet connectivity.

b) Discuss different types of ISPs.

(9 Marks)

ii) Describe how redundancy is achieved in Internet infrastructure. (3 Marks)

iii) Define a Content Delivery Network (CDN) in one sentence. (3 Marks)

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Rajagiri College of Social Sciences (Autonomous)

Continuous Assessment Examination - II

October 2024

I MCA

Code: MCA104 **Total Time: 60 minutes**

Sub: Data Communications and Computer Networks **Total Mark: 20**

Level	Levels of Learning
L1	Remembering
L2	Understanding
L3	Applying
L4	Analyzing
L5	Evaluating
L6	Creating

SECTION A

Each question has 10 marks

Sl.no	Question	CO Mapped	Bloom's Taxonomy level
1 A	A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x3+1. a. What is the actual bit string transmitted? b. Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?	MCA104(3).1	L3
	OR		
1 B	Explain the sliding window protocols in detail.	MCA104(3).1	L2
	SECTION B *Each question has 10 marks*		
2A	Explain how ARP can be used to resolve communication failure between two devices due to incorrect MAC address mapping in a network.	MCA104(3).2	L3
	OR		
2B	Make a comparison between LLC and MAC in detail.	MCA104(3).2	L2

Name:
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RCSS
RAJAGIRI COLLEGE OF SOCIAL SCIENCES

Rajagiri College of Social Sciences (Autonomous)

Continuous Assessment Examination - 1

September 2024

I MCA

Code: MCA104 **Total Time: 90 minutes**

Sub: Data Communications and Computer Networks **Total Mark: 40**

Level	Levels of Learning
L1	Remembering
L2	Understanding
L3	Applying
L4	Analyzing
L5	Evaluating
L6	Creating

SECTION A

Each question has 10 marks

Sl.no	Question	CO Mapped	Bloom's Taxonomy level
1	Explain the polar coding schemes of digital signals. – **Embedded Question**	**MCA104(1).1**	**L2**
2	Write notes on: 1. Multiplexing 2. Hierarchical Layering 3. Encapsulation/Decapsulation 4. PDU		
	OR		
3	What is Pulse Code Modulation (PCM), and how does it convert analog signals into digital form for transmission?		

SECTION B

Each question has 10 marks

4	How does wireless transmission differ from wired transmission, and what are the key factors that affect the quality and range of wireless signals? – **Embedded Question**	**MCA104(2).2**	**L2**
5	What is a Public Switched Telephone Network (PSTN), and how does it enable voice communication between different telephones?		

P.T.O

	OR		
6	Write short notes on the following types of transmission media: 1. HTTP 2. SMTP 3. UDP 4. ICMP		